A Candidate Proof of Finite Image for Rank-Three Surface-Group Representations in Genus at Least Five

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Version 0.1.3-candidate
Expanded review copy
10 August 2026

Abstract

We give a candidate proof that a complex rank-three representation of an orientable genus-g punctured surface group, whose conjugacy class has finite mapping-class-group orbit, has finite image whenever $g \geq 5$. Landesman–Litt prove this under the strict inequality $r < \sqrt{g+1}$, which treats rank three directly only for $g \geq 9$. The proof combines a corrected endpoint estimate, explicit Harder–Narasimhan analysis in genera six and five, universal-extension and boundary-cocycle obstructions for adjoint bundles, and a rank-three-specific formal-propagation argument. Separate spectral-projector and canonical-Deligne determinant reconstructions are retained for auditability but are not needed for the main elimination. The manuscript is unrefereed, and the assembled argument has not received independent mathematical review.

Status and scope. This is an unrefereed candidate proof. The published inputs are recorded in section 2. No claim is made in genus three or four.

Contents

1 Statement and proof architecture

Let $\Sigma_{g,n}$ be an orientable genus-g surface with n punctures. The mapping class group $\mathrm{Mod}_{g,n}$ acts on conjugacy classes of representations of $\pi_1(\Sigma_{g,n})$ by precomposition.

Theorem 1.1 (Main theorem)**.** *Let $g \geq 5$ and $n \geq 0$. If*

$$\rho : \pi_1(\Sigma_{g,n}) \longrightarrow \mathrm{GL}_3(\mathbf{C})$$

has finite $\mathrm{Mod}_{g,n}$-orbit up to conjugacy, then ρ has finite image.

Landesman–Litt prove finite image when

$$r < \sqrt{g+1},$$

with strict inequality [4, Theorem 1.2.1]. Thus their theorem treats $r = 3$ directly for $g \geq 9$. The new cases are $g = 8, 7, 6, 5$.

The proof has eight layers.

(1) At the square endpoint $r^2 - 1 = g$, equality in the Landesman–Litt bundle estimate is shown to be impossible.

(2) A universal-extension construction produces large trivial subbundles of quotient normalizers. A Cech boundary-cocycle identity rules out such a subbundle once it has generic rank at least three and the full extension boundary is injective.

(3) In genus five, proper positive-dimensional projective closures reduce to adjoint constituents of ranks $5+3$ or $6+2$; the rank-six monomial constituent is treated separately.

(4) The dense rank-eight genus-five case is reduced to a finite HN branch tree.

(5) Every no-high-slope branch is eliminated by Brill–Noether, Kirillov, relative-line, boundary-cocycle, block-vanishing, or exact jet arguments. In particular, after closed descent the zero-weight $q = 9$ branch supplies a trivial quotient-normalizer subbundle of rank at least three.

(6) The sole high-HN branch is another instance of the same obstruction: the universal extension supplies a trivial rank-four subbundle, while parabolic stability makes the full boundary injective.

(7) For independent reconstruction, the former residual $q = 9$ configuration is also eliminated by a spectral projector, and an integral periodic-chain calculation extends the minimal-nilpotent determinant character without poles for arbitrary contact order.

(8) Projective-extension uniqueness descends residual adjoint invariants from any dominant versal base to a finite etale cover, and a direct small-extension devissage propagates their vanishing through every Artin thickening when the unitary anchor is irreducible. Reducible unitary anchors are treated separately by the fibral-unitary isotypic decomposition and an exact coefficient/multiplicity inequality. The remaining rigidity, integrality, nonabelian-Hodge, and socle-induction steps of [4] then apply.

2 Published inputs and conventions

We use the following statements exactly in the forms indicated.

2.1 Landesman–Litt

From [4]:

- Theorem 1.2.1: finite image for an MCG-finite rank-r representation when $r < \sqrt{g+1}$, with arbitrary punctures.
- Proposition 2.1.1: semisimple subquotients preserve MCG-finiteness.
- Lemma 2.2.3: an irreducible MCG-finite representation admits a *projective* globalization on a scheme finite etale over $\mathcal{M}_{g,n}$.
- Proposition 2.3.4 and Corollary 2.3.5: a finite-determinant linear lift exists after a dominant etale base change. We never use this dominant lift as though it were a finite cover.
- Lemmas 2.1.4–2.1.5, Lemma 2.2.3, and the proof of Lemma 2.2.2: finite-index image of a dominant versal base, geometric existence of the projective globalization, and uniqueness of the projective intertwiners defining it. We do not attribute a separate uniqueness statement to Lemma 2.2.3.
- Lemmas 2.3.2, 2.4.1, and 2.4.2: finite determinant, extension of unitarity/finiteness from a normal fibre subgroup, and the published general fibral-unitary Artin decomposition. Direct adjoint devissage treats an irreducible unitary anchor; Lemma 2.4.2 is essential for reducible anchors.

- Theorem 1.7.1 and Lemma 6.1.1: a nonzero sub-local system of $R^1\pi_*U$ has rank at least $2g - 2\,\mathrm{rk}\,U$, and a low-rank sub-local system produces a Hodge vector with controlled period-map rank.

- Proposition 5.2.3 and Theorem 6.2.1: the parabolic multiplication estimate and the published strict Artin vanishing.

- Sections 8.2–8.7: strong rigidity, integrality, unitary finiteness, nonabelian-Hodge constancy, and characteristic-socle induction.

From [5] we use Lemma 6.2.3 and Proposition 6.3.6, including the inequalities appearing in their proofs after quotienting the high HN part.

2.2 Small-slope Brill–Noether theory

From Brambila-Paz–Grzegorczyk–Newstead [1] we use:

- the vector-bundle Clifford estimate;

- the generated-subbundle conclusion in the low-slope equality range;

- the necessary inequality
$$n \le d + (n-k)g$$
for a semistable bundle of rank n, degree $0 < d \le n$, and at least k sections.

We apply these estimates separately to HN graded pieces.

2.3 Sections of universal curves

Earle–Kra prove that for genus at least three the Teichmuller curve with n marked points has exactly the n tautological holomorphic sections [3]; for $n = 0$ there are none. Chen–Salter prove that the Birman exact sequence does not virtually split for $g \ge 4$ [2].

2.4 Canonical Deligne and coparabolic convention

For a unitary local system, E_* denotes the parabolic bundle associated with the Deligne canonical extension, with weights in $[0, 1)$. The parabolic non-GGG estimate is applied to

$$P_* = (E_*)^\vee \otimes \omega_C(D),$$

where $\omega_C(D)$ carries the trivial parabolic structure. The ordinary bundle that enters all section counts is its coparabolic degree-zero piece.

Proposition 2.1 (Underlying ordinary bundle)**.** *With the preceding convention,*

$$\widehat{P}_0 = E^\vee \otimes \omega_C.$$

Consequently
$$\deg(E^\vee \otimes \omega_C) = \mathrm{rk}(E)(2g-2) + S,$$
where $S = -\deg E$ is the total canonical parabolic weight.

Proof. Choose $0 < \varepsilon$ smaller than every positive weight. By the coparabolic definition, $\widehat{P}_0 = P_\varepsilon$. The parabolic tensor rule and the trivial parabolic structure on $\omega_C(D)$ give

$$P_\varepsilon = (E_*^\vee)_\varepsilon \otimes \omega_C(D) + (E_*^\vee)_0 \otimes \omega_C.$$

The parabolic dual formula used in [5, Section 2] is

$$(E_*^\vee)_\alpha = (\widehat{E}_{-\alpha})^\vee(-D).$$

For small ε, periodicity gives $\widehat{E}_{-\varepsilon} = E$. Hence the first summand is

$$E^\vee(-D) \otimes \omega_C(D) = E^\vee \otimes \omega_C.$$

Locally at a marked point, the second summand is a submodule of this first summand: the factor $(-D)$ in the dual formula shifts precisely the positive-weight directions. Thus their sum is $E^\vee \otimes \omega_C$.

Moreover $\operatorname{pardeg}((E_*)^\vee) = 0$, while the trivial parabolic line $\omega_C(D)$ has parabolic degree $2g - 2 + n$. Therefore the input to [5, Proposition 6.3.6] lies exactly in its boundary case, with no hidden marked-point divisor in the ordinary bundle. $\square$

This identity also fixes the section deficit used below. If the Landesman–Litt multiplication map has rank zero, every section of $E^\vee \otimes \omega_C$ lies in the contraction kernel, so the deficit δ in the proof of [5, Lemma 6.2.3] is zero.

3 Reduction to actual adjoint constituents

A reducible semisimple rank-three representation has constituents of rank at most two and is already in the published strict range for $g \geq 5$. We may therefore treat an irreducible rank-three system V.

Proposition 3.1 (Projective closure alternatives)**.** *If the projective image of V is infinite, the irreducible adjoint constituents are multiplicity-free and occur in one of the following forms:*

projective closure	$\operatorname{ad} V$
PGL_3 *-dense*	U_8
$\mathrm{PGL}_2 \simeq \mathrm{SO}_3$	$U_5 \oplus U_3$
monomial, component quotient S_3	$U_6 \oplus U_2$
monomial, component quotient C_3	$U_3^+ \oplus U_3^- \oplus \chi \oplus \chi^{-1}$.

The only coefficient ranks not already in the published strict range at genus five are therefore 8, 6, *and* 5, *and each of those occurs once.*

Proof. The connected reductive projective closure acts irreducibly on a three-dimensional space. If its connected semisimple part is nontrivial, it is either SL_3 in the standard representation or SL_2 through Sym^2. These give respectively $\mathfrak{sl}_3$ and $\mathrm{Sym}^4 \oplus \mathrm{Sym}^2$, with multiplicity one.

If the connected component is a torus, irreducibility gives three distinct weights permuted transitively by a quotient C_3 or S_3. The connected projective torus is the full diagonal torus: its rational character space is a nonzero invariant subspace of the irreducible two-dimensional standard permutation module. Hence the six root characters are distinct. For S_3 they form one orbit, giving an irreducible rank-six off-diagonal module, while the traceless diagonal part is the irreducible rank-two standard module. For C_3 the roots split into the two oriented three-cycles, giving nonisomorphic rank-three modules U_3^+, U_3^-; the diagonal part splits into the two distinct nontrivial characters. All constituents are therefore multiplicity-free. Dimension three is prime, so there is no nontrivial tensor-product case. $\square$

The table records the actual residual coefficient systems that must be excluded on finite etale moduli covers. Formal propagation will not assume that the finite-cover geometric contradiction persists after the dominant etale base changes of [4, Lemma 2.4.2]. Instead, theorems 14.1 and 14.2 below works with the whole adjoint: projective uniqueness pulls its residual local system back from a finite cover, and a direct small-extension induction handles the Artin thickness.

4 The fixed-part interface

Lemma 4.1 (Closed descent in the dense zero-weight branch)**.** *Let $B \to \mathcal{M}_{g,n}$ be a connected smooth finite-etale scheme cover carrying the canonical projective globalization of an irreducible unitary, projectively MCG-finite rank-three system. Let $\pi : X \to B$ be the compact universal curve, let $D = \sum_i s_i(B)$, and put $X^\circ = X \setminus D$. Suppose that the fibral projective monodromy factors through $\pi_1(\Sigma_g)$ and is Zariski dense in PGL_3. Let $\mathcal{U}$ be its adjoint local system on X°, and assume that $\mathcal{U}$ is unitary on the total space, as ensured in the application by theorem 4.3. Then*

$$H^0\Big(B, R^1\pi_*\mathcal{U}\Big) \xrightarrow{\sim} H^0\Big(B, R^1\pi^\circ_*\mathcal{U}\Big)\,.$$

Moreover, every nonzero fixed vector on the right, together with the rank-zero Hodge line supplied by the fixed-part construction, descends to a connected finite-etale scheme cover of $\mathcal{M}_g$.

Proof. The projective puncture monodromies are trivial, so $\mathcal{U}$ extends uniquely across every marked divisor. The relative residue sequence is

$$0 \longrightarrow R^1\pi_*\mathcal{U} \longrightarrow R^1\pi^\circ_*\mathcal{U} \longrightarrow \bigoplus_i s_i^*\mathcal{U}(-1) \longrightarrow R^2\pi_*\mathcal{U}. \tag{1}$$

In fact the last term vanishes because dense adjoint monodromy has no fibrewise invariants, but only exactness is needed below.

Write $\Gamma = \pi_1(B)$ and let K_i be the Birman point-pushing subgroup for the ith mark. Let $\mathrm{ev}_i : K_i \to \pi_1(\Sigma_g)$ be evaluation followed by filling the other marks. The subgroup $\Gamma \cap K_i$ has finite index in K_i. Under the canonical projective globalization, point-pushing by α acts on $s_i^*\mathcal{U}$ as

$$\mathrm{Ad}(\bar\rho(\mathrm{ev}_i(\alpha)))\,.$$

The lift of α along the boundary section s_i moves the universal-curve basepoint together with the ith marked point. After filling the marks, this diagonal lift acts on the based closed surface group by conjugation with $\mathrm{ev}_i(\alpha)$. The intertwiner calculation in the proof of [4, Lemma 2.2.2] therefore gives the displayed adjoint action on $s_i^*\mathcal{U}$. This is a direct diagonal/filling computation; Lemma 2.2.3 is used only for the geometric existence of the globalization. After filling the marked points, the evaluation image of $\Gamma \cap K_i$ has finite index in $\pi_1(\Sigma_g)$. Its image under $\bar\rho$ is therefore still Zariski dense: a finite-index subgroup of a dense subgroup of the connected group PGL_3 is dense. Consequently

$$H^0(B, s_i^*\mathcal{U}) \subseteq \mathfrak{sl}_3^{\overline{\mathrm{Ad}\,\bar\rho(\mathrm{ev}_i(\Gamma\cap K_i))}} = 0.$$

Taking global flat sections in (1) proves the displayed isomorphism.

Let $\bar\Gamma$ be the image of Γ under the forgetful map to Mod_g; it has finite index. On closed twisted cohomology, a forgotten point-push acts by an inner automorphism together with the corresponding coefficient conjugation, and hence acts trivially on group cohomology. Explicitly, if z is a cocycle and the push is represented by a, the transformed cocycle is $g \mapsto z(g) + (g-1)z(a)$, differing from z by a coboundary. Thus the monodromy action on $R^1\pi_*\mathcal{U}$ factors through $\bar\Gamma$. The factored projective representation is stabilized by $\bar{\Gamma}$. Its dense image has trivial centralizer in PGL_3, so the uniqueness proof of [4, Lemma 2.2.2] constructs the canonical projective globalization over the corresponding finite cover of $\mathcal{M}_g$ even if the closed projective representation has no global GL_3-lift. After passage to a common finite cover, its pullback to the pointed family is the original globalization: both are defined by the same unique projective intertwiners from the proof of [4, Lemma 2.2.2]. Their adjoint local systems are therefore identified. The fixed vector descends to the closed cover. Intersecting with a level subgroup makes it a scheme cover. The closed adjoint globalization has trivial determinant and irreducible unitary fibral restriction. The exact sequence of [4, Lemma 2.1.5] makes the fibre group normal, so [4, Lemma 2.4.1(2)] makes the closed adjoint unitary on the total space. Apply [4, Theorem 4.1.1] to $\mathcal{U} \oplus \mathcal{U}^\vee$, which has the natural real structure noted there, and take the $\mathcal{U}$-summand. Applying the same theorem with $D = \varnothing$ identifies its closed F^1 term with $\pi_*(E \otimes \omega_\pi)$. For the open family, trivial puncture monodromy identifies the weight-one term with $R^1\pi_*\mathcal{U}$, and strictness of the inclusion of variations of mixed Hodge structure gives

$$F^1 R^1 \pi^\circ_* \mathcal{U} \cap R^1 \pi_* \mathcal{U} = F^1 R^1 \pi_* \mathcal{U}.$$

Hence the rank-zero Hodge representative lies in

$$H^0(C, E \otimes \omega_C),$$

with no logarithmic residue, and descends with the fixed vector. $\square$

Lemma 4.2 (A rank-one fixed part gives multiplication rank zero)**.** *Let U be a unitary local system on the total space of a punctured versal family over a smooth quasi-projective complex base whose classifying map is etale. If*

$$H^0(B, R^1\pi^\circ_* U) \neq 0,$$

then, after possibly replacing U by its complex conjugate, there is a nonzero

$$v \in H^0(C, E \otimes \omega_C(D))$$

for which the Landesman–Litt multiplication map

$$H^0(C, E^\vee \otimes \omega_C) \longrightarrow H^0(C, \omega_C^2(D))$$

has rank zero.

Proof. A nonzero flat section spans a trivial irreducible rank-one sub-local system $L \subset R^1\pi^\circ_* U$. By [4, Lemma 6.1.1], after possible complex conjugation there is a nonzero $v \in L_b \cap F^1\mathcal{H}_b$ whose period derivative has rank at most $\operatorname{rk} L/2 = 1/2$. Its rank is an integer, so it is zero. Equivalently, in the proof of that lemma the two-Hodge-type case cannot occur in rank one, and the remaining case has identically zero period map. The Hodge–de Rham degeneration used in [4, Theorem 4.1.1] identifies F^1 with the logarithmic one-form term. Because the classifying map is etale, the cotangent map c_b^* in [4, Theorem 5.1.6] is an isomorphism, so the factorization in that theorem identifies zero period derivative with the displayed zero multiplication map. $\square$

Lemma 4.3 (Compatibility for the new adjoint coefficients)**.** *After a finite etale base change fixing the projective-closure data, every coefficient of rank 8, 6, or 5 used below is a globally defined irreducible unitary local system on the total family, and complex conjugation does not alter its parabolic HN problem.*

Proof. The dense adjoint U_8 is self-dual by the trace form. The principal PGL_2 constituents U_5 and U_3 are the self-dual representations Sym^4 and Sym^2. The S_3 monomial constituent U_6 is self-dual under the pairing of E_{ij} with E_{ji}. For a unitary system, complex conjugation is the dual, so these coefficients are self-conjugate. The two oriented rank-three constituents in the cyclic monomial case are exchanged by conjugation, but both already lie in the published strict range.

The determinants of U_8, U_5, U_3 are trivial. On U_6 the product of the six torus weights t_i/t_j is one and the component group contributes only a finite permutation determinant. Thus every new coefficient has finite determinant. Its fibral restriction is irreducible and unitary, so [4, Lemma 2.4.1] makes the total coefficient unitary. Finite etale base change preserves versality. Hence theorem 4.2 applies on every finite cover required later. □

5 The endpoint rank-one lemma

Proposition 5.1 (Strict stable pairing estimate)**.** *Let C have genus $g \geq 2$, let D be reduced, and let E_* be parabolically stable of parabolic degree zero and rank $d \geq 2$. For a nonzero*

$$v \in H^0(C, E \otimes \omega_C(D)),$$

let s be the rank of the multiplication map of [4, Proposition 5.2.3]

$$H^0(C, E^\vee \otimes \omega_C) \longrightarrow H^0(C, \omega_C^2(D))$$

Then

$$\boxed{d \geq g - s + 1.}$$

Proof. The claim is automatic if $s \geq g$, so assume $s < g$. The published weak estimate gives $d \geq g - s$. If the displayed stronger estimate failed, then $d = g - s$.

Put $F = E^\vee \otimes \omega_C$, let K be the corank-one contraction kernel, and let A be the saturation in F of the evaluation image of $H^0(K)$. Saturation stays inside the saturated kernel K, and every section of either K or A factors through A. Hence, with $\delta = h^0(F) - h^0(A)$,

$$\delta = h^0(F) - h^0(K) = s.$$

If $c = d - \mathrm{rk}\, A$, Proposition 6.3.6 of [5] gives

$$d \geq gc - s.$$

Since $c \geq 1$ and $d + s = g$, equality forces $c = 1$.

Let N^t be the HN part of A with slope $> 2g - 2$. The proof of the same proposition, using $H^1(N^t) = 0$ to preserve the deficit s, gives

$$\mathrm{rk}(F/N^t) \geq g - s = d.$$

Since $\mathrm{rk}\, F = d$, $N^t = 0$. The intermediate inequality of [5, Lemma 6.2.3] is

$$\mu(F) + 2 \leq (2g - \mu(F))d + 2s.$$

The quotient-slope lemma gives $\mu(F) \geq 2g - 2$. At $\mu(F) = 2g - 2$ the two sides are equal because $d + s = g$, and increasing $\mu(F)$ increases the left side while decreasing the right. Therefore equality is forced. If S is the total parabolic weight, then

$$\mu(F) = 2g - 2 + \frac{S}{d},$$

and therefore $S = 0$. The bundle E is now ordinary stable of degree zero, and F is stable of slope $2g-2$: all weights lie in $[0,1)$, so total weight zero forces every weight to be zero and the strictly increasing parabolic flag to be the trivial one-step flag.

We can now return to the complete inequality chain in the proof of [5, Lemma 6.2.3]. Since $N^t = 0$, the bundle A satisfies its Harder–Narasimhan hypotheses, and that proof gives

$$\deg F + (1-g)d \le \frac{\deg A}{2} + \operatorname{rk} A + s \le \frac{\mu(F)\operatorname{rk} A}{2} + \operatorname{rk} A + s.$$

Here $\mu(F) = 2g-2$, $\operatorname{rk} A = d-1$, and $d+s = g$. Both extreme terms are therefore equal to $d(g-1)$:

$$\deg F + (1-g)d = d(g-1)$$

and

$$\frac{\mu(F)(d-1)}{2} + (d-1) + s = g(d-1) + s = d(g-1).$$

Thus equality holds throughout. In particular,

$$\deg A = \mu(F)\operatorname{rk} A,$$

so $\mu(A) = \mu(F)$. But $d \ge 2$ and $c = 1$, hence A is a nonzero proper subbundle of the ordinary stable bundle F. Stability requires $\mu(A) < \mu(F)$, a contradiction.

This last step uses no additional Clifford theorem: the contradiction is already contained in the equality conditions of the Landesman–Litt inequality chain. The condition $d \ge 2$ is exactly what ensures $\operatorname{rk} A = d - 1 > 0$. □

Remark 5.2 (The rank-one exception)**.** The restriction $d \ge 2$ is sharp. If $D = \varnothing$ and $E \in \operatorname{Pic}^0(C)$ is nontrivial, then $h^0(E\omega_C) = h^0(E^{-1}\omega_C) = g-1$. Multiplication by any nonzero $v \in H^0(E\omega_C)$ is injective on $H^0(E^{-1}\omega_C)$, so $s = g-1$ and $d+s = g$. Thus the published weak bound is attained in rank one.

Corollary 5.3 (Strict fixed-part estimate)**.** *Let U be an irreducible unitary fibral local system of rank d on a punctured versal genus-g family. Every nonzero irreducible sub-local system $L \subset R^1\pi_*U$ has*

$$\operatorname{rk} L \ge \begin{cases} 2g-2, & d = 1, \\ 2g-2d+2, & d \ge 2. \end{cases}$$

Proof. For $d = 1$ use [4, Theorem 1.7.1]. For $d \ge 2$, by [4, Lemma 6.1.1], after possible complex conjugation there is a nonzero Hodge vector whose multiplication rank s is at most $\operatorname{rk} L/2$. The proposition gives $s \ge g-d+1$, and the claim follows. □

Corollary 5.4 (Artin vanishing through rank g)**.** *Let $g \ge 3$. In the setup of [4, Theorem 6.2.1], the conclusion remains valid when $\operatorname{rk}_A \mathcal{V} \le g$.*

Proof. A possible term $U \otimes \pi^*W$ satisfies

$$uw \le g$$

and, if it contributes a nonzero invariant, the preceding fixed-part lower bound is at most w. If $w = 1$, that lower bound is at least two for every $u \le g$. If $w \ge 2$ and $u = 1$, then

$$uw = w \ge 2g-2 > g.$$

If $w \geq 2$ and $u \geq 2$, then $u \leq g/2$ and

$$uw \geq 2u(g-u+1) \geq 2g > g.$$

Both alternatives contradict $uw \leq g$. $\square$

Theorem 5.5 (Square-endpoint finite-image theorem)**.** *Let $g \geq 3$, $n \geq 0$, and $r^2 \leq g+1$. Every MCG-finite representation*

$$\rho : \pi_1(\Sigma_{g,n}) \longrightarrow \mathrm{GL}_r(\mathbf{C})$$

has finite image.

Proof. Only the boundary case $r^2 = g+1$ is new. Then $\mathrm{rk}(\mathrm{ad}\,\rho) = r^2 - 1 = g$, so theorem 5.4 replaces [4, Theorem 6.2.1] in the proofs of strong cohomological rigidity [4, Proposition 8.2.1] and formal constancy [4, Lemma 8.5.1]. The projective and linear integrality arguments, unitary finiteness, and the semisimple nonabelian-Hodge deformation in [4, Sections 8.3–8.5] then apply unchanged. The PVHS-to-unitary input is used in the larger range $r < 2\sqrt{g+1}$, which is automatic here.

It remains to check the non-semisimple step at equality. In the notation of [4, Lemma 8.6.1], let the characteristic semisimple factors have ranks a, b. Then

$$ab \leq \frac{(a+b)^2}{4} \leq \frac{g+1}{4}.$$

Writing $\rho_2^\vee \otimes \rho_1 = \bigoplus_i \sigma_i^{\oplus n_i}$ gives $n_i \dim \sigma_i \leq (g+1)/4$ and $\dim \sigma_i < g$. Any nonzero projected extension support has, by [4, Theorem 7.2.1], rank at least

$$2g - 2\dim \sigma_i \geq \frac{3g-1}{2} > \frac{g+1}{4} \geq n_i \dim \sigma_i,$$

a contradiction. Thus every characteristic extension splits. The socle induction of [4, Section 8.7] completes the proof. $\square$

This includes the square endpoints $(g, r) = (3, 2), (8, 3), (15, 4), (24, 5), \ldots$. The detailed endpoint proof is logically independent of the genus-seven, six, and five branch analysis below.

6 The first super-endpoint

The next proposition moves one rank above the stable endpoint. It is used only when the multiplication rank is zero, but it replaces two separate low-genus calculations.

Proposition 6.1 (First super-endpoint pairing lemma)**.** *Let $g \geq 3$, let D be reduced, and let E_* be parabolically stable of parabolic degree zero and rank*

$$d = g + 1.$$

Write S for the total canonical parabolic weight of E_*. Suppose $S \neq 1$. Then no nonzero

$$v \in H^0(C, E \otimes \omega_C(D))$$

can have multiplication rank zero.

Proof. Assume the multiplication map is zero and put $F = E^\vee \otimes \omega_C$. Let $K \subset F$ be the contraction kernel and let $A \subset K$ be the saturation of the evaluation image. Then $h^0(A) = h^0(F)$. If

$$c = d - \operatorname{rk} A \geq 1,$$

the proof of [5, Proposition 6.3.6] gives $d \geq gc$. Since $g + 1 < 2g$, one has $c = 1$ and hence $\operatorname{rk} A = g$.

Let $N \subset A$ be the HN part of slope $> 2g - 2$. The same proof, after quotienting by N, gives $\operatorname{rk}(F/N) \geq g$, so $\operatorname{rk} N \leq 1$. If $\operatorname{rk} N = 1$, then $\operatorname{rk}(F/N) = g$ and the complete inequality chain in [5, Lemma 6.2.3] is

$$2g \leq \mu(F/N) + 2 \leq (2g - \mu(F/N))g \leq 2g.$$

Thus equality holds throughout,

$$\mu(F/N) = \mu(A/N) = 2g - 2.$$

Since

$$\deg F = (g + 1)(2g - 2) + S,$$

this forces

$$\deg N = 2g - 2 + S, \qquad \deg(A/N) = (g - 1)(2g - 2),$$

and therefore $\deg(F/A) = 2g - 2$. If $M \subset E$ is the saturated line generated by v, contraction gives

$$F/A \simeq M^\vee \otimes \omega_C,$$

so $\deg M = 0$. Its induced parabolic weights are nonnegative, contradicting parabolic stability. Hence $N = 0$.

The intermediate inequality now gives

$$\mu(F) + 2 \leq (2g - \mu(F))(g + 1).$$

Using

$$\mu(F) = 2g - 2 + \frac{S}{g + 1}$$

yields

$$S \leq \frac{2(g + 1)}{g + 2} < 2.$$

The parabolic-degree identity makes S a nonnegative integer. Since $S \neq 1$, it follows that $S = 0$. Thus E is an ordinary stable zero-degree bundle and F is stable of slope $2g - 2$. Since $g + 1 > 1$, $H^0(E) = 0$, and Riemann–Roch gives

$$h^0(F) = (g + 1)(g - 1) = g^2 - 1.$$

The generically globally generated bundle A has rank g and every HN slope lies in $[0, 2g - 2)$. The BPGN Clifford estimate and stability give

$$2g^2 - 2g - 2 \leq \deg A < 2g^2 - 2g,$$

so

$$\deg A \in \{2g^2 - 2g - 2,\ 2g^2 - 2g - 1\}.$$

Set $G = A^\vee \otimes \omega_C$. Every HN slope of G is positive, and Serre duality gives

$$(\deg G, h^0(G)) = (1, g) \quad \text{or} \quad (2, g + 1).$$

The BPGN small-slope bound, applied HN-piece by HN-piece, gives at most $g - 1$ sections in either total degree. This is a contradiction. $\square$

Proposition 6.2 (Classification of the weight-one boundary)**.** *Under the hypotheses of Proposition 6.1, suppose instead that a rank-zero vector exists. Then necessarily $S = 1$, and if $M \subset E$ is its saturated Hodge line and $A \subset E^\vee \otimes \omega_C$ its saturated evaluation kernel, then*

$$A \simeq \omega_C^{\oplus g}, \qquad \deg M = -1,$$

and the underlying bundle is the universal extension

$$0 \longrightarrow M \longrightarrow E \longrightarrow \mathcal{O}_C^{\oplus g} \longrightarrow 0.$$

Equivalently, the boundary map $\mathbf{C}^g \to H^1(C, M)$ is an isomorphism.

Proof. The proof of Proposition 6.1 gives $S < 2$ and excludes $S = 0$, so $S = 1$. Parabolic stability gives $H^0(E) = 0$, and hence

$$h^0(F) = \chi(F) = (g+1)(g-1) + 1 = g^2.$$

The rank-g bundle A carries all these sections and has every HN slope in $[0, 2g-2]$. Clifford and the HN upper bound give

$$g^2 \leq \frac{\deg A}{2} + g, \qquad \deg A \leq g(2g-2),$$

so $\deg A = 2g(g-1)$. Thus $G = A^\vee \otimes \omega_C$ has degree zero, nonnegative HN slopes, and

$$h^0(G) = h^1(A) = g.$$

A semistable degree-zero bundle with as many sections as its rank is trivial, so $G \simeq \mathcal{O}_C^g$ and $A \simeq \omega_C^g$.

Since $\deg F = (g+1)(2g-2) + 1$, one has $\deg(F/A) = 2g - 1$. The contraction identity $F/A \simeq M^\vee \otimes \omega_C$ gives $\deg M = -1$. Tensoring the contraction sequence by ω_C^{-1} and dualizing gives

$$0 \to M \to E \to \mathcal{O}_C^g \to 0.$$

The boundary map $\mathbf{C}^g \to H^1(M)$ is injective because $H^0(E) = H^0(M) = 0$, and both spaces have dimension g. Hence it is an isomorphism. □

Proposition 6.3 (Normalizer boundary-cocycle obstruction)**.** *Let C be a smooth projective connected complex curve. Let E be a vector bundle of Lie algebras whose generic fibre is $\mathfrak{sl}_3$, let $M \subset E$ be a saturated line, and put*

$$G = E/M, \qquad K = N_E(M)/M \subset G.$$

Let

$$\partial : H^0(C, G) \longrightarrow H^1(C, M)$$

be the boundary map. If ∂ is injective, there is no vector subspace $U \subset H^0(C, K)$ of dimension at least three for which

$$U \otimes \mathcal{O}_C \longrightarrow K$$

has generic rank $\dim U$*.*

Proof. Since M is saturated, G is locally free. The quotient bracket identifies

$$K = \ker\bigl(G \longrightarrow G \otimes M^{-1}\bigr) = N_E(M)/M.$$

In particular, K is saturated in G. The normalizer is a Lie subalgebra, M is an ideal in it, and hence K is a sheaf of Lie algebras.

The action of the normalizer on M induces the grading character

$$\lambda : K \longrightarrow \mathrm{End}(M) = \mathcal{O}_C, \qquad [x, m] = \lambda(x)m.$$

For a global section x of K, the function $\lambda(x)$ is constant because C is projective.

Let $x, y \in H^0(C, K)$. On an affine cover choose local lifts $X_i, Y_i \in N_E(M)$ and write

$$a_{ij} = X_j - X_i \in M, \qquad b_{ij} = Y_j - Y_i \in M.$$

With the corresponding Cech convention, these cocycles represent ∂x and ∂y. Since $[M, M] = 0$,

$$[X_j, Y_j] - [X_i, Y_i] = \lambda(x)b_{ij} - \lambda(y)a_{ij}.$$

Consequently

$$\partial[x, y] = \lambda(x)\partial y - \lambda(y)\partial x.$$

Injectivity of ∂ gives the identity

$$[x, y] = \lambda(x)y - \lambda(y)x. \tag{2}$$

Thus every vector subspace of $H^0(C, K)$ is bracket-closed.

Suppose now that U has generic rank at least three. Replacing it by a three-dimensional subspace, let $F = \mathbb{C}(C)$ and let $U_F \subset K_F$ be its three-dimensional generic fibre. The quotient boundary law (2) extends F-bilinearly to U_F. The quotient normalizer of a nonzero element of $\mathfrak{sl}_3$ has dimension $1, 2, 3$, or 4 in the regular nonnilpotent, regular nilpotent, semisimple $(2, 1)$, or minimal-nilpotent cases, respectively. Hence only the last two cases remain.

In the semisimple $(2, 1)$ case,

$$K_F \simeq \mathfrak{sl}_2(F), \qquad \lambda = 0.$$

Since $\dim_F U_F = 3$, one has $U_F = K_F$, whereas (2) makes it abelian, a contradiction.

In the minimal-nilpotent case, after extending F if necessary, take $m = E_{12}$. Modulo Fm, the quotient normalizer has basis

$$D = \mathrm{diag}(1/2, -1/2, 0), \quad H = \mathrm{diag}(1/2, 1/2, -1), \quad P = E_{13}, \quad Q = E_{32},$$

with

$$\lambda(D) = 1, \qquad \lambda(H) = \lambda(P) = \lambda(Q) = 0$$

and

$$[D, P] = \tfrac{1}{2}P, \quad [D, Q] = \tfrac{1}{2}Q, \quad [H, P] = \tfrac{3}{2}P, \quad [H, Q] = -\tfrac{3}{2}Q, \quad [P, Q] = 0.$$

If $\lambda|_{U_F} = 0$, then $U_F = \ker\lambda = \langle H, P, Q\rangle_F$, but this is not abelian. If $\lambda|_{U_F} \neq 0$, choose $X \in U_F$ with $\lambda(X) = 1$ and put $V = U_F \cap \ker\lambda$. Equation (2) gives

$$[V, V] = 0, \qquad [X, v] = v \quad (v \in V).$$

If two independent vectors $u = \alpha H + \beta P + \gamma Q$ and $v = \alpha' H + \beta' P + \gamma' Q$ commute, their P- and Q-coefficients give

$$\alpha\beta' - \alpha'\beta = 0, \qquad \alpha\gamma' - \alpha'\gamma = 0.$$

If either α or α' were nonzero, these equations would make u, v proportional. Thus the only two-dimensional abelian subspace of $\langle H, P, Q\rangle_F$ is $\langle P, Q\rangle_F$. Writing

$$X = D + cH + aP + bQ,$$

the eigenvalues of ad_X on P, Q are

$$\frac{1+3c}{2}, \qquad \frac{1-3c}{2},$$

which cannot both equal one. This is again a contradiction. $\square$

Proposition 6.4 (Adjoint universal-extension obstruction)**.** *Let C have genus $g \geq 5$. Let E be an adjoint bundle with generic fibre $\mathfrak{sl}_3$, let $M \subset E$ be a saturated line of degree -1, put $L = M^{-1}$ and $G = E/M$, and suppose that*

$$W \otimes \mathcal{O}_C \subset G, \qquad \dim W = g.$$

Let $Q = G/(W \otimes \mathcal{O}_C)$ and let $P \subset E$ be the inverse image of $W \otimes \mathcal{O}_C$. Assume $H^0(P) = 0$, $h^0(Q \otimes L) \leq h$, and that bracket with M descends to

$$\beta : G \longrightarrow G \otimes L.$$

Then

$$\mathcal{O}_C^{g-h} \subset \ker \beta.$$

Consequently, the configuration is impossible if $g - h \geq 5$. If in addition $H^0(C, E) = 0$, it is impossible already when $g - h \geq 3$.

Proof. Riemann–Roch gives $h^1(M) = g$. The boundary map of

$$0 \to M \to P \to W \otimes \mathcal{O}_C \to 0$$

is injective because $H^0(P) = 0$, hence is an isomorphism $e : W \xrightarrow{\sim} H^1(M)$. Let $\phi : W \to H^0(Q \otimes L)$ be the quotient component of β and put $W_0 = \ker \phi$, so $\dim W_0 \geq g - h$.

If $h^0(L) = 0$, the remaining block $W_0 \otimes \mathcal{O}_C \to W \otimes L$ vanishes. If $L = \mathcal{O}_C(r)$, write that block as sA, where s is the unique section of L and $A : W_0 \to W$ is constant. The morphism-of-extensions identity says that the composite

$$W_0 \xrightarrow{A} W \xrightarrow{e} H^1(M) \xrightarrow{s_*} H^1(\mathcal{O}_C)$$

is zero. The last map is an isomorphism by $0 \to \mathcal{O}_C(-r) \to \mathcal{O}_C \to k(r) \to 0$, so $A = 0$. Thus $W_0 \otimes \mathcal{O}_C \subset \ker \beta$.

For a nonzero $m \in \mathfrak{sl}_3$, the quotient-adjoint kernel has dimension $4, 3, 2,$ or 1 according as m is minimal nilpotent, semisimple of type $2+1$, regular nilpotent, or regular semisimple. Therefore a kernel of rank at least five is impossible. If $H^0(C, E) = 0$, the boundary map $H^0(C, G) \to H^1(C, M)$ is injective. Proposition 6.3, applied to $\mathcal{O}_C^{g-h} \subset \ker \beta$, gives the stronger conclusion when $g - h \geq 3$. $\square$

Corollary 6.5 (Two rank-three applications)**.** *The rank-eight dense adjoint coefficient in genus seven and the rank-six off-diagonal monomial coefficient in genus five have no rank-one fixed part.*

Proof. For the dense adjoint in genus seven, Proposition 6.1 reduces a possible rank-zero vector to the weight-one universal extension of Proposition 6.2. There $G = E/M \simeq \mathcal{O}_C^7$, so Proposition 6.4 applies with $h = 0$ and gives an impossible rank-seven quotient-bracket kernel.

For the rank-six off-diagonal monomial coefficient in genus five, $d = 6 = g + 1$ and the exact local totals are

$$0, \quad \frac{3}{2}, \quad 2, \quad \frac{5}{2}, \quad 3.$$

The global total is integral and therefore cannot equal one. Apply Proposition 6.1. $\square$

7 The rank-eight coefficient in genera seven and six

7.1 Genus seven

The Artin coefficient optimization leaves only ranks seven and eight with base multiplicity one. Rank seven is covered by Corollary 5.3, while rank eight is excluded by Corollary 6.5. No separate genus-seven HN branch calculation is needed.

7.2 Genus six

Here F has rank eight, $2g-2=10$, and $\deg F = 80+S$. A rank-one fixed part gives a saturated line $M \subset E$ and a rank-seven evaluation kernel $A \subset F$ with

$$F/A \simeq M^\vee \otimes \omega_C, \qquad q := \deg(F/A) = 10 - \deg M \geq 11.$$

Let $N \subset A$ be the HN part of slope > 10, and write $t = \operatorname{rk} N$, $x = \deg N$, and $y = x - S$. The quotient-rank estimate gives $t \leq 2$.

If $t=2$, then $\operatorname{rk}(F/N) = g = 6$, so equality holds throughout the Landesman–Litt chain. Consequently

$$\mu(F/N) = \mu(A/N) = 10, \qquad \deg(F/A) = 10,$$

and hence $\deg M = 0$, contrary to parabolic stability.

Suppose $t=1$ and put $B = A/N$. The two quotient-slope bounds give

$$10 \leq \mu(F/N) = \frac{80-y}{7} \leq \frac{12 \cdot 7 - 2}{8} = \frac{41}{4},$$

so $y \in \{9, 10\}$. Since

$$\operatorname{rk} B = 6, \qquad \deg B = 80 - q - y,$$

the HN upper degree $\deg B \leq 60$ and the Clifford inequality, together with $q \geq 11$ above, give

$$q + y \geq 20, \qquad q \leq y + 2, \qquad q \geq 11.$$

Let $G = B^\vee \otimes \omega_C \subset E/M$ and let $P \subset E$ be its inverse image. The two offsets and all allowed values of q are therefore

y	q	$\deg P$	$\deg(E/P)$
9	11	-1	$1-S$
10	$11, 12$	0	$-S.$

The row $y=10$ contradicts parabolic stability because the proper subbundle P has ordinary degree zero and nonnegative induced parabolic weights. For $y=9$, the inequality $x = S+9 > 10$ gives $S \geq 2$. Let β be the total parabolic weight of the quotient line E/P, and let m be the number of nonscalar punctures. Stability of P would require

$$-1 + (S - \beta) < 0, \qquad \text{hence} \qquad \beta > S - 1.$$

At a scalar puncture the adjoint parabolic structure is trivial, so its quotient-line weight is zero. At each nonscalar puncture that weight is strictly less than one, while every nonscalar adjoint puncture contributes total weight 2 or 3. Consequently

$$\beta < m \leq \frac{S}{2} \leq S - 1,$$

a contradiction. Thus no high-HN branch survives.

With $N = 0$, the exact no-high list is

$$(S, q) = (0, 11), (0, 12), (0, 13), (0, 14), (2, 12).$$

For the zero-weight cases put $G = E/M$. The relevant BPGN bounds, applied to all positive-degree HN factors, are

q	$\deg G$	$h^0(G)$	maximum allowed
11	1	6	6
12	2	7	6
13	3	8	7
14	4	9	8.

Hence only $(S, q) = (0, 11)$ remains among the zero-weight branches. There the rank-seven degree-one quotient G has a saturated trivial evaluation subbundle $W = \mathcal{O}_C^6$. Its quotient Q is a line, $\deg M = -1$, and $h^0(Q \otimes M^{-1}) \leq 1$ on a nonhyperelliptic fibre. The inverse image $P \subset E$ has $H^0(P) = 0$ by stability. Proposition 6.4, with $g = 6$ and $h = 1$, therefore produces an impossible rank-five quotient-bracket kernel.

For $(S, q) = (2, 12)$ the same low-degree equality analysis gives

$$E/M \simeq \mathcal{O}_C^7, \qquad \deg M^{-1} = 2.$$

The ordinary $\mathfrak{sl}_3$ fibre and the contracted

$$\mathfrak{gl}_2 \ltimes (\mathbf{C}^2 \oplus (\mathbf{C}^2)^\vee)$$

fibre have no one-dimensional ideals, so the quotient bracket is fibrewise nonzero. On a nonhyperelliptic genus-six curve, a degree-two line has at most one section, and the matrix has a common zero. Contradiction.

We have therefore obtained the required adjoint Artin vanishing for every rank-three unitary system in genus at least six.

8 Genus five: monomial coefficient

Let U_6 be the irreducible off-diagonal constituent of an irreducible monomial rank-three unitary system. Its rank is

$$6 = g + 1$$

in genus five. The exact local total weights are

$$0, \quad \frac{3}{2}, \quad 2, \quad \frac{5}{2}, \quad 3.$$

The global total is integral because $\deg E + S = 0$, and hence cannot equal one. Proposition 6.1 excludes a rank-one fixed part. The rank-five orthogonal constituent is covered by Corollary 5.3, and the smaller constituents lie in the published strict range. Thus every proper positive-dimensional projective closure is eliminated in genus five.

9 Genus-five dense HN reconstruction

Let E_* be the parabolically stable rank-eight adjoint Deligne bundle. Put

$$F = E^\vee \otimes \omega_C, \qquad \deg F = 64 + S, \qquad h^0(F) = 32 + S.$$

A rank-one fixed part gives a saturated line $M \subset E$ and a rank-seven kernel $A \subset F$ with

$$F/A \simeq M^\vee \otimes \omega_C, \qquad q := \deg(F/A) = 8 - \deg M \geq 9.$$

Let N be the HN part of A with slope > 8, and put $t = \operatorname{rk} N$, $x = \deg N$, $B = A/N$. The proof of [5, Proposition 6.3.6] and Clifford give

$$64 - (8-t)\frac{10(8-t)-2}{9-t} \leq x - S \leq 8t, \tag{7.1}$$

$$x \geq S + 10t + q - 14, \tag{7.2}$$

$$x \geq 8t + 1. \tag{7.3}$$

The integral offsets are

t	$x - S$
1	$5, 6, 7, 8$
2	$15, 16$
3	$24.$

For $t = 3$, (7.2) gives $q \leq 8$. For $t = 2$, the offset 16 produces a proper degree-zero rank-six subbundle of E, while offset 15 forces

$$t = 2, \quad q = 9, \quad x = S + 15, \quad S \geq 2, \quad (A/N)^\vee \otimes \omega_C \simeq \mathcal{O}_C^5. \tag{7.4}$$

For $t = 1$, put $y = x - S$, $B = A/N$, and $G = B^\vee \otimes \omega_C \subset E/M$, and let $P \subset E$ be the inverse image of G. The HN upper degree and Clifford inequalities give

$$q + y \geq 16, \qquad q \leq y + 4, \qquad q \geq 9.$$

Moreover

$$\deg P = y - 8, \qquad \deg(E/P) = 8 - y - S.$$

Thus the complete integer table is

y	allowed q	$\deg P$	extra consequence of $x = S + y > 8$
5	$\varnothing$	-3	$--$
6	10	-2	$S \geq 3$
7	$9, 10, 11$	-1	$S \geq 2$
8	$9, 10, 11, 12$	0	$--.$

The row $y = 8$ contradicts parabolic stability because P has degree zero and nonnegative induced weights. For $y = 6, 7$, let β be the total weight of the quotient line E/P and let m be the number of nonscalar punctures. Stability of P requires

$$\beta > S - 2 \quad (y = 6), \qquad \beta > S - 1 \quad (y = 7).$$

At a scalar puncture the adjoint parabolic structure is trivial, so its quotient-line weight is zero. Every nonscalar quotient-line weight is strictly less than one and every nonscalar adjoint puncture contributes total weight 2 or 3, so $\beta < m \leq S/2$. This immediately contradicts the $y = 7$ inequality. For $y = 6$ it also contradicts stability when $S \geq 4$. If $S = 3$, there is exactly one full-flag puncture, so $\beta < 1$, whereas stability requires $\beta > 1$. Hence every $t = 1$ row is impossible, and (7.4) is the only high-HN family.

If $N = 0$, the intermediate inequality gives

$$S \leq \frac{16}{3}, \qquad \max(9, 8+S) \leq q \leq 14 - S.$$

Since an adjoint puncture contributes 0, 2, or 3, the exact list is

$$\begin{aligned} S = 0 &: q = 9, 10, 11, 12, 13, 14, \\ S = 2 &: q = 10, 11, 12, \\ S = 3 &: q = 11. \end{aligned}$$

For

$$G = A^\vee \otimes \omega_C \simeq E/M$$

one has

$$\operatorname{rk} G = 7, \qquad \deg G = q - 8 - S, \qquad h^0(G) = q - 4.$$

For the zero-weight $S = 0$ branches, applying BPGN HN-piece by HN-piece gives

q	$\deg G$	$h^0(G)$	maximum allowed
9	1	5	5
10	2	6	6
11	3	7	6
12	4	8	7
13	5	9	8
14	6	10	8.

Thus only $q = 9, 10$ survive. In the punctured branch $(S, q) = (2, 12)$ the rank-seven degree-two bundle has eight sections, whereas the same calculation, now allowing degree-zero HN factors, gives a maximum of seven. The no-high residual list is

$$(0, 9), (0, 10), (2, 10), (2, 11), (3, 11).$$

10 Zero-weight genus-five branches

10.1 The branch $(S, q) = (0, 10)$

Here $G = E/M$ has rank seven, degree two, and six sections, while $L = M^{-1} = \det G$ has degree two. The trace form $\operatorname{tr}(XY)$ (equivalently, a nonzero scalar multiple of the Killing form) and the bracket define the Kirillov form

$$\Omega : \Lambda^2 G \longrightarrow L.$$

For $0 \neq m \in \mathfrak{sl}_3$, the rank of ad_m is four or six. The radical of the induced form on $\mathfrak{sl}_3/\mathbf{C}m$ is the centralizer of m modulo $\mathbf{C}m$, so the form has the same rank. Thus an isotropic subspace in the seven-dimensional quotient has dimension at most five.

The positive integral HN degrees of G sum to two. The BPGN small-slope bound shows that six sections are possible only in the following two cases:

(i) G is semistable of rank seven and degree two;

(ii) there is an exact sequence $0 \to J \to G \to Q \to 0$ with $(\mathrm{rk}, \deg, h^0)(J) = (1, 1, 1)$ and $(\mathrm{rk}, \deg, h^0)(Q) = (6, 1, 5)$, and every section of Q lifts.

In case (i), coprimality makes G stable. Let V be the saturation of the evaluation image of its six sections. Since V is generically generated, its HN quotients have nonnegative degree. Stability and the degree-one BPGN bound force $\mathrm{rk}\, V = 6$ and $\deg V = 0$: if $\deg V = 1$, equality of the section count would give a positive degree-one HN line in V, which would destabilize G. Hence the evaluation determinant has no zeros and

$$V \simeq \mathcal{O}_C^6.$$

Since $h^0(L) \le 1$ on a nonhyperelliptic genus-five curve, all entries of $\Omega|_V$ vanish on a common fibre (or vanish identically), producing a six-dimensional isotropic subspace.

In case (ii), let V be the saturation of the evaluation image of $H^0(Q)$. Semistability of Q of slope $1/6$ gives $\deg V = 0$, while five sections force $\mathrm{rk}\, V = 5$ and

$$V \simeq \mathcal{O}_C^5.$$

Pulling V back to G gives $0 \to J \to P \to \mathcal{O}_C^5 \to 0$. All five sections of Q lift, so this extension splits:

$$P \simeq J \oplus \mathcal{O}_C^5.$$

Let $T = Q/\mathcal{O}_C^5$, a degree-one line. Determinants give $L = J \otimes T$. The form entries on P lie in $H^0(L)$ on $\Lambda^2 \mathcal{O}_C^5$ and in $H^0(T)$ on $J \otimes \mathcal{O}_C^5$. If $h^0(T) = 0$, all cross terms vanish, while the $H^0(L)$ entries have a common zero (or vanish identically). If $h^0(T) = 1$, the product of the unique sections of J and T spans $H^0(L)$ whenever the latter is nonzero; the zero of the section of T kills every form entry. In either case a fibre of P is six-dimensional isotropic, a contradiction.

10.2 The branch $(S, q) = (0, 9)$

Here $S = 0$ says that the projective puncture monodromies are trivial; by itself it does not say that $n = 0$. Since this is the dense adjoint branch, theorem 4.1 moves the fixed vector and its Hodge line to a finite cover of $\mathcal{M}_g$. We run the entire branch on that closed family. All constructions below use only the adjoint Lie bundle and its associated Azumaya algebra, so no global GL_3-lift is required.

Thus $E = \mathrm{ad}\, \bar{V}$ is stable of degree zero, $M \subset E$ is saturated with $\deg M = -1$, and

$$G = E/M$$

is stable of rank seven, degree one, and has five sections. The saturated evaluation image is

$$W = \mathcal{O}_C^5 \subset G,$$

with rank-two quotient $R = G/W$, $\deg R = 1$, and $\det R \simeq L := M^{-1}$. Let

$$\beta : G \longrightarrow G \otimes L, \qquad K = \ker \beta.$$

The inverse image of W in E is the universal extension of $\mathcal{O}_C^5$ by M. If

$$h = h^0(R \otimes L),$$

then stability and the small-slope bounds give $h \le 2$, and the universal-extension identity gives

$$\mathcal{O}_C^{5-h} \subset K. \tag{3}$$

Since E is stable of degree zero, $H^0(C, E) = 0$, so the boundary map $H^0(C, G) \to H^1(C, M)$ is injective. Because $h \le 2$, (3) supplies a generically injective trivial subbundle of rank at least three in K, contradicting Proposition 6.3. This eliminates the zero-weight $q = 9$ branch.

10.3 Independent determinant and spectral reconstruction

For auditability, we retain a second elimination of the former $h = 2$ configuration. It is not used in the proof above. Suppose $h = 2$, so $\mathcal{O}_C^3 \subset K$. The following proposition supplies the determinant and degree reduction used in this independent reconstruction.

Proposition 10.1 (Determinant reduction in the $h = 2$ branch)**.** *Let $Z \subset E$ be the saturated centralizer of the Hodge line. One of the following occurs.*

(i) *$L = M^{-1}$ is effective of degree one;*

(ii) *the Hodge line is minimal nilpotent on every fibre, the image of ad_m is saturated, and the grading character $\lambda : K \to \mathcal{O}_C$ is surjective and nonzero on the displayed $\mathcal{O}_C^3 \subset K$.*

Proof. At the generic point, K is the normalizer of the Hodge line modulo that line. The inclusion $\mathcal{O}_C^3 \subset K$ therefore rules out every regular element of $\mathfrak{sl}_3$: its quotient normalizer has dimension one in the nonnilpotent case and dimension two in the regular-nilpotent case. The remaining nonzero orbit types are the semisimple type $(2, 1)$ and the minimal nilpotent type.

The invariant trace form identifies $Z^\perp$ with the dual quotient of E. The L-valued Kirillov form

$$\kappa_m : \Lambda^2(E/Z) \longrightarrow L, \qquad \kappa_m(\bar{x}, \bar{y}) = \mathrm{tr}(m[x, y]),$$

is generically symplectic, since both the semisimple $(2, 1)$ orbit and the minimal nilpotent orbit in $\mathfrak{sl}_3$ have dimension four. Its Pfaffian is therefore a generically nonzero morphism

$$\det(E/Z) \longrightarrow L^2.$$

Because $\det E \simeq \mathcal{O}_C$, there is a unique effective divisor T such that

$$\det Z \simeq M^2 \otimes \mathcal{O}_C(T). \tag{4}$$

The divisor T is the degeneracy divisor of the integral orbit form; it vanishes exactly when $\mathrm{im}(\mathrm{ad}_m)$ is saturated in $Z^\perp \otimes L$. Indeed, under the trace identification $Z^\perp \simeq (E/Z)^\vee$, the determinant of

$$\overline{\mathrm{ad}}_m : E/Z \longrightarrow Z^\perp \otimes L$$

is the square of the displayed Pfaffian. Its divisor is therefore $2T$, and it is a unit at every point exactly when the image is saturated.

Suppose first that the generic element is semisimple of type $(2, 1)$. Then the generic line normalizer equals the centralizer. More intrinsically, bracket with M defines a global grading morphism

$$\lambda : K \longrightarrow \mathcal{O}_C, \qquad [x, m] = \lambda(x)m,$$

whose kernel is Z/M. At the generic point $\mathrm{tr}(m^2) \neq 0$, while trace invariance gives $\lambda(x)\,\mathrm{tr}(m^2) = 0$. Thus λ vanishes generically and hence identically. Consequently the equality extends across every degeneration point and

$$K = Z/M, \qquad \det K \simeq M \otimes \mathcal{O}_C(T).$$

The inclusion $\mathcal{O}_C^3 \subset K$ has full generic rank. Since K is a rank-three subsheaf of the stable bundle G of rank seven and degree one, $\deg K \leq 0$. Taking determinants of the inclusion gives $\deg K \geq 0$. Thus $\deg K = 0$, $\deg T = 1$, and a nonzero section trivializes $\det K$. Equation (4) then gives $L \simeq \mathcal{O}_C(T)$, so L is effective.

Now suppose the generic element is minimal nilpotent. Minimal nilpotence holds on every fibre: the equations $m^2 = 0$ and $\operatorname{rk} m \leq 1$ define a closed locus, and the saturated line never vanishes. Let

$$\lambda : K \longrightarrow \mathcal{O}_C, \qquad [x, m] = \lambda(x)m,$$

be the grading character and let Z_λ be the effective divisor defined by $\operatorname{im} \lambda = \mathcal{O}_C(-Z_\lambda)$. Its kernel is Z/M, and (4) gives

$$\det K \simeq M \otimes \mathcal{O}_C(T - Z_\lambda). \tag{5}$$

If λ vanishes on $\mathcal{O}_C^3$, then $\mathcal{O}_C^3 \subset Z/M$ has full generic rank. Stability again gives $\deg(Z/M) \leq 0$, while the determinant of the inclusion gives $\deg(Z/M) \geq 0$. Hence $\det(Z/M) \simeq \mathcal{O}_C$, and (4) yields $\deg T = 1$ and $L \simeq \mathcal{O}_C(T)$.

If λ is nonzero on $\mathcal{O}_C^3$, its restriction is a nonzero map from a trivial bundle to $\mathcal{O}_C(-Z_\lambda)$. Since Z_λ is effective and C is projective, this forces $Z_\lambda = 0$ and makes the restriction surjective. The integral determinant theorem of section 13 gives a nonzero morphism $\det K \to M$. Combining it with (5) produces a nonzero morphism $\mathcal{O}_C(T) \to \mathcal{O}_C$, and hence $T = 0$. Thus the adjoint image is saturated, and alternative (ii) holds. □

In alternative (i), theorem 10.4 gives a section of the universal curve over the finite cover of $\mathcal{M}_g$ supplied above, contradicting Chen–Salter. It remains only to eliminate alternative (ii).

Proposition 10.2 (Spectral-projector obstruction)**.** *Suppose the Hodge line is minimal nilpotent on every fibre, K has rank four, and the grading character*

$$\lambda : K \longrightarrow \mathcal{O}_C, \qquad [x, m] = \lambda(x)m,$$

is surjective and nonzero on the displayed $\mathcal{O}_C^3 \subset K$. Then the configuration is impossible.

Proof. Put $C_0 = \ker \lambda$. Then C_0 is the quotient centralizer $\mathfrak{z}(m)/M$, of rank three, and

$$0 \longrightarrow C_0 \longrightarrow K \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

The determinant morphism gives $\deg C_0 = \deg K \leq -1$. Etale-locally a generator of the saturated minimal-nilpotent line is $m = E_{12}$. Modulo M, the centralizer then has the basis

$$E_{13}, \quad E_{32}, \quad H = \operatorname{diag}(1/2, 1/2, -1).$$

Its derived algebra is the span of E_{13}, E_{32}, a saturated rank-two subbundle. The trace form has precisely this radical and $\operatorname{tr}(H^2) = 3/2$, so it descends perfectly to the quotient line. Thus the derived algebra

$$R_0 = [C_0, C_0]$$

has rank two, and for $T = C_0/R_0$ one has $T^{\otimes 2} \simeq \mathcal{O}_C$ and $\deg T = 0$.

Let

$$U = \mathcal{O}_C^3 \cap C_0 \simeq \mathcal{O}_C^2.$$

If the projection $U \to T$ vanishes, then $U \subset R_0$ and determinants give a nonzero map $\mathcal{O}_C \to \det R_0$, contradicting $\deg \det R_0 = \deg C_0 \leq -1$. Otherwise $T \simeq \mathcal{O}_C$. Choose a global section $u \in H^0(C, U)$ mapping to a generator of T and a global section $x \in H^0(C, \mathcal{O}_C^3)$ with $\lambda(x) = 1$. Set $x_t = x + tu$.

Choose locally a generator m of M and a lift X_t of x_t to E. Then $[X_t, m] = m$. Any other lift is $X_t + fm$, and

$$X_t + fm = \exp(-fm)X_t \exp(fm)$$

because $m^2 = 0$. Hence the characteristic polynomial of X_t is independent of the lift and has global holomorphic, therefore constant, coefficients.

In a frame with $m = E_{12}$, every lift has the form

$$\operatorname{diag}(1/2, -1/2, 0) + aE_{12} + bE_{13} + cE_{32} + d\,\operatorname{diag}(1/2, 1/2, -1).$$

The section u changes d by a nonzero affine function of t. Fix one point of C and choose t so that its value D is zero there. The three eigenvalues are

$$\frac{D+1}{2}, \qquad \frac{D-1}{2}, \qquad -D,$$

At the chosen point the spectrum is therefore $\{1/2, -1/2, 0\}$. Since the characteristic coefficients are global holomorphic functions, this is the spectrum at every point; in particular, $1/2, -1/2$ are the unique ordered pair differing by one.

Let $\alpha = 1/2$, $\beta = -1/2$, and $\gamma = 0$. The projector onto the γ-eigenspace is

$$P_\gamma = \frac{(X_t - \alpha I)(X_t - \beta I)}{(\gamma - \alpha)(\gamma - \beta)}.$$

It satisfies $P_\gamma m = mP_\gamma = 0$. The conjugacy formula above makes it independent of the choice of lift, while ordinary frame changes conjugate it as an endomorphism. The local projectors therefore glue to a global rank-one idempotent in the Azumaya algebra associated with $\bar{V}$. Its trace-free part is a nonzero section of the adjoint bundle E, contradicting $H^0(C, E) = 0$. ☐

This completes the independent $h = 2$ reconstruction.

10.4 Fixed-part Hodge section and relative degree-one contradiction

Lemma 10.3 (Fixed-part rank-zero Hodge section)**.** *Let $\pi^\circ : \mathcal{C}^\circ \to B$ be a punctured versal family over a smooth quasi-projective complex base whose classifying map is etale, and let $\mathcal{U}$ be unitary. If*

$$H^0(B, R^1\pi^\circ_*\mathcal{U}) \neq 0,$$

then, after complex conjugation if necessary, there is a global fibrewise nonzero section

$$v \in H^0(B, \pi_*(\mathcal{E} \otimes \omega_\pi(D)))$$

whose Landesman–Litt multiplication map has rank zero on every fibre. Equivalently, adjunction gives a global map

$$\omega_\pi(D)^{-1} \longrightarrow \mathcal{E}.$$

Proof. A nonzero fixed vector spans a trivial irreducible rank-one local system $\mathbb{L} \subset R^1\pi^\circ_*\mathcal{U}$. For a general complex unitary coefficient, apply [4, Proposition 4.2.2] to $\mathbb{W} = R^1\pi^\circ_*\widetilde{\mathcal{U}}$ and $\widetilde{\mathbb{L}} = \mathbb{L}$; the trivial line has its natural real structure. The proof of [4, Lemma 6.1.1] then splits into two cases. Its second case would give the real rank-one system $\widetilde{\mathbb{L}} = \mathbb{L}$ two conjugate Hodge types, which is impossible in rank one. Thus, after conjugating $\mathcal{U}$ if necessary, the first case gives a global morphism of variations from a constant line into $F^1R^1\pi^\circ_*\mathcal{U}$. Its period map is identically zero.

The filtered de Rham description and Hodge–de Rham degeneration in [4, Theorem 4.1.1] give

$$F^1R^1\pi^\circ_*\mathcal{U} \simeq \pi_*(\mathcal{E} \otimes \omega_\pi(D)),$$

Because the classifying map is etale, the cotangent map c_b^* in [4, Theorem 5.1.6] is an isomorphism, so the factorization in that theorem identifies zero second fundamental form with zero multiplication map. Thus the corresponding global section has rank-zero multiplication on every fibre. The nonzero map from the irreducible local system $\mathbb{L}$ is injective, so its Hodge-bundle image is fibrewise nonzero. □

Proposition 10.4 (Relative Hodge-line closure). *Let $\pi : X \to B$ be the pullback of the universal curve to a connected finite-etale scheme cover, with B integral and regular. Let $\mathcal{E}$ be a vector bundle on X, and suppose a line bundle $\mathcal{H}$ on B gives a generically nonzero morphism*

$$\pi^*\mathcal{H} \otimes \omega_\pi(D)^{-1} \longrightarrow \mathcal{E}.$$

Let $\mathcal{S}$ be the saturation of its rank-one image in $\mathcal{E}$ and put $\mathcal{M} = \mathcal{S}^{**}$. Then:

(i) *$\mathcal{M}$ is invertible and the induced map $\mathcal{M} \hookrightarrow \mathcal{E}$ is injective;*

(ii) *on the generic fibre, $\mathcal{M}_\eta$ is exactly the saturated line generated by the Hodge vector;*

(iii) *if $\deg(\mathcal{M}_\eta) = -1$, then $\mathcal{L} = \mathcal{M}^{-1}$ has relative degree one on all of B;*

(iv) *if $\mathcal{L}_\eta$ is effective, then π admits a section.*

Proof. The total space X is regular. The sheaf $\mathcal{M}$ is coherent reflexive of rank one. Since every regular local ring is a UFD, a rank-one reflexive module on X is invertible. Double dualizing the inclusion $\mathcal{S} \hookrightarrow \mathcal{E}$ gives $\mathcal{M} \to \mathcal{E}$; it is generically injective, and its kernel would be a torsion subsheaf of the line bundle $\mathcal{M}$, hence is zero.

The quotient $\mathcal{M}/\mathcal{S}$ is supported in codimension at least two. It does not meet the generic fibre: a closed point of the generic curve has codimension one in X. Therefore

$$\mathcal{M}_\eta = \mathcal{S}_\eta,$$

and the right side is saturated by construction. Fibre degree of a line bundle is locally constant in a connected smooth proper family, proving (iii). The base twist $\mathcal{H}$ has fibre degree zero.

Assume now that $\mathcal{L}_\eta$ is effective. Upper semicontinuity makes

$$\{b \in B : h^0(X_b, \mathcal{L}_b) \geq 1\}$$

a closed subset containing the generic point, hence all of B. A degree-one line bundle on a positive-genus smooth curve has at most one independent section, so $h^0(X_b, \mathcal{L}_b) = 1$ for every b; Riemann–Roch also makes $h^1(X_b, \mathcal{L}_b) = g - 1$ constant. Cohomology and base change therefore makes $\pi_*\mathcal{L}$ invertible and identifies its fibres with $H^0(X_b, \mathcal{L}_b)$.

The evaluation

$$\pi^*\pi_*\mathcal{L} \longrightarrow \mathcal{L}$$

is nonzero on every fibre. It cuts out an effective Cartier divisor $Z \subset X$. Its restriction to each fibre is the unique degree-one zero divisor. Since the section remains a nonzerodivisor on every smooth integral fibre, Z is a relative effective Cartier divisor and is flat over B. It is proper with zero-dimensional fibres, hence finite. Thus $Z \to B$ is finite locally free of degree one, therefore an isomorphism. Its inverse is a section of π. □

Apply the proposition on the connected finite-etale base on which the fixed part has been globalized. Lemma 10.3 supplies the global evaluation, and its generic saturation is the line M from the fibre calculation. Thus a fibrewise proof that M^{-1} is effective produces an actual section of that universal curve. In the zero-weight effective subcases of section 10.2, theorem 4.1 first replaces the pointed base by a finite cover of $\mathcal{M}_g$; only there does the resulting section contradict Chen–Salter's non-virtual-splitting theorem for $g \geq 4$ [2]. No line defined only on an HN stratum is being extended, and no arbitrary pointed section is declared contradictory.

For a nontrivial rank-one sub-local system rather than an actual fixed vector, [4, Proposition 4.2.2 and Lemma 6.1.1] supply the same evaluation with a flat line bundle pulled back from the base. In pointed branches a section is not itself contradictory; Earle–Kra identifies it with one of the tautological marked sections [3].

11 Punctured no-high branches

11.1 The branch $(S, q) = (2, 11)$

Let p be the unique $2 + 1$ puncture. Here G has rank seven, degree one, seven sections, and nonnegative HN slopes. The degree-one BPGN bound shows that equality is possible only with a positive degree-one HN line J and a rank-six degree-zero quotient carrying six sections. The quotient is $\mathcal{O}_C^6$, every quotient section lifts, and the extension splits. Thus

$$G \simeq J \oplus \mathcal{O}_C^6, \qquad \deg J = 1, \qquad h^0(J) = 1, \qquad \deg L = 3.$$

The degree-one line globalizes on the full finite cover: intrinsically $J = \det G$, because the complementary evaluation summand has trivial determinant. Generic effectivity and the degree-one Abel closed immersion give a global section, and Earle–Kra forces $J \simeq \mathcal{O}_C(r)$ for a marked point r. Since $\det E \simeq \mathcal{O}_C(-2p)$,

$$L \simeq \mathcal{O}_C(2p + r).$$

The four blocks of the quotient bracket have coefficients in

$$\mathcal{O}(2p + r), \quad \mathcal{O}(2p), \quad \mathcal{O}(2p + 2r), \quad \mathcal{O}(2p + r).$$

On a very general pointed genus-five fibre, each indicated section space is one-dimensional and every unique section vanishes at p. Hence the bracket vanishes at p, making M_p a one-dimensional ideal of

$$\mathfrak{gl}_2 \ltimes (\mathbf{C}^2 \oplus (\mathbf{C}^2)^\vee),$$

which has no such ideal.

11.2 The branch $(S, q) = (3, 11)$

Here $G \simeq \mathcal{O}_C^7$ and there is one full-flag puncture p. The canonical adjoint lattice has determinant $\mathcal{O}_C(-3p)$, while $\det G = \det E \otimes L$ is trivial. Hence

$$L \simeq \mathcal{O}_C(3p).$$

Choose the very general marked fibre so that p is non-Weierstrass; then $h^0(\mathcal{O}_C(3p)) = 1$, and every bracket coefficient vanishes to order at least three at p.

The full-flag contracted fibre has exactly three line ideals:

$$\mathbf{C}E_{12}, \qquad \mathbf{C}E_{23}, \qquad \mathbf{C}E_{31}.$$

For each, the bracket with the opposite root has exact order one:

$$[E_{12}, E_{21}] = zH_1, \quad [E_{23}, E_{32}] = zH_2, \quad [E_{31}, E_{13}] = -z(H_1 + H_2).$$

These pure-root equalities also control an arbitrary saturated lift. Let e_α be the canonical root frame of one of the three ideals and $e_{-\alpha}$ its opposite frame. A primitive generator has the form $m = e_\alpha + zr + O(z^2)$; multiplying by a unit removes the e_α component of r. Since $[e_\alpha, e_{-\alpha}] = zH_\alpha$, while a bracket with $e_{-\alpha}$ has Cartan component only from the e_α component, one gets

$$\mathrm{pr}_{\mathfrak{t}}[m, e_{-\alpha}] = zH_\alpha \pmod{z^2\mathfrak{t}_R}.$$

Thus the order-one Cartan term cannot cancel for any saturated lift. A matrix of sections of $\mathcal{O}(3p)$ would vanish to order at least three. Contradiction.

11.3 The branch $(S, q) = (2, 10)$

Here

$$0 \to \mathcal{O}_C^6 \to G \to D \to 0, \qquad \deg D = 0, \qquad \deg L = 2,$$

and $D = L(-2p)$. If $h^0(L) = 0$, then $h^0(G \otimes L) \le 1$, whereas the ordinary quotient-adjoint map has rank at least three and restricts to rank at least two on a hyperplane. Thus L is effective. Its unique degree-two divisor can be supported only at the sole nonordinary point:

$$L \simeq \mathcal{O}_C(2p), \qquad D \simeq \mathcal{O}_C.$$

The six global bracket sections then have length-two jets spanning at most one complex dimension.

Let $W(z)$ be the local rank-six subbundle. A one-dimensional jet image gives a five-dimensional kernel in the constant section space. Because the global subbundle is trivial, evaluation identifies it with a five-dimensional subspace of W_0. For a moving vector $v(z) = v_0 + zv_1$, the vanishing jet equation is $B_0v_0 = 0$ and $B_1v_0 + B_0v_1 = 0$; projection to $\operatorname{coker} B_0$ removes the moving-frame term B_0v_1. Hence the induced map $T : \ker B_0 \to \operatorname{coker} B_0$ must vanish on a five-dimensional subspace.

We now display the finite local calculation. Put $R = \mathbf{C}[[z]]$ and use the canonical $2 + 1$ lattice frame

$$(e_1, \dots, e_8) = (H_1, H_2, E_{12}, E_{21}, zE_{13}, zE_{23}, E_{31}, E_{32}),$$

where $H_1 = \mathrm{diag}(1, -1, 0)$ and $H_2 = \mathrm{diag}(0, 1, -1)$. The special fibre is $\mathfrak{gl}_2 \ltimes (V \oplus V^\vee)$. A line with nonzero Levi part has quotient-bracket rank at least three. For a radical vector (u, v), the ranks at most two are exactly the pure case, in which one of u, v vanishes, and the mixed isotropic case $u, v \ne 0$, $v(u) = 0$.

In the pure normal form take

$$m = e_5 + z\sum_{i=1}^{8} a_ie_i + O(z^2).$$

On $G_0 = \mathfrak{p}_R/\langle e_5\rangle$, the constant bracket has $B_0(e_4) = -e_6$ and vanishes on the other quotient basis vectors. With ordered bases

$$\ker B_0 = (e_1, e_2, e_3, e_6, e_7, e_8), \qquad \operatorname{coker} B_0 = (e_1, e_2, e_3, e_4, e_7, e_8),$$

the induced first-order map is

$$T_{\text{pure}} = \begin{pmatrix} a_1 & a_1 & -a_4 & 0 & 1 & 0 \\ a_2 & a_2 & 0 & 0 & 1 & 0 \\ -a_3 & 2a_3 & 2a_1 - a_2 & 0 & 0 & 1 \\ 3a_4 & 0 & 0 & 0 & 0 & 0 \\ 2a_7 & 2a_7 & 0 & 0 & -a_1 - a_2 & -a_4 \\ 0 & 3a_8 & a_7 & 0 & -a_3 & a_1 - 2a_2 \end{pmatrix}.$$

The R-linear Lie-lattice involution

$$X \longmapsto -DX^{\mathrm{t}}D^{-1}, \qquad D = \mathrm{diag}(z, z, 1),$$

exchanges e_5 with $-e_7$ and e_6 with $-e_8$, so this calculation covers both pure radical orbits, including arbitrary first-order corrections.

In the mixed isotropic normal form take

$$m = e_5 + e_8 + z\sum_{i=1}^{8} a_i e_i + O(z^2).$$

Here $B_0(e_2) = 3e_8$ and $B_0(e_4) = -e_6 + e_7$. Use

$$\ker B_0 = (e_1, e_3, e_6, e_7, e_8), \qquad \mathrm{coker}\, B_0 = (e_1, e_2, e_3, e_4, e_6),$$

where $e_7 = e_6$ and $e_8 = 0$ in the cokernel. Then

$$T_{\text{mixed}} = \begin{pmatrix} a_1 & -a_4 & 0 & 1 & 0 \\ a_2 & 0 & -1 & 1 & 0 \\ -a_3 & 2a_1 - a_2 & 0 & 0 & 1 \\ 3a_4 & 0 & 0 & 0 & 0 \\ 2a_6 + 2a_7 & 0 & -a_1 + 2a_2 & -a_1 - a_2 & -a_4 \end{pmatrix}.$$

The respective constant submatrices are

$$(T_{\text{pure}})_{\{1,3\},\{5,6\}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad (T_{\text{mixed}})_{\{1,2,3\},\{3,4,5\}} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Both have determinant one. Changing the moving frame replaces B_1 by $B_1 + B_0U - VB_0$, and rescaling the line adds a multiple of B_0; neither operation changes the induced map $T : \ker B_0 \to \mathrm{coker}\, B_0$. Thus T has rank at least two in the pure case and at least three in the mixed case for every first-order correction. A one-dimensional two-jet image would force T to vanish on a five-dimensional subspace, impossible.

12 The final high-HN branch

The sole remaining branch is

$$t = 2, \quad q = 9, \quad \deg N = S + 15, \quad S \geq 2, \quad (A/N)^\vee \otimes \omega_C \simeq \mathcal{O}_C^5.$$

Put

$$G = E/M, \qquad L = M^{-1}, \qquad W = \mathcal{O}_C^5 \subset G, \qquad Q = G/W.$$

Then

$$\deg M = -1, \quad \deg L = 1, \quad \mathrm{rk}\, Q = 2, \quad \deg Q = 1 - S, \quad h^0(Q \otimes L) \leq 1.$$

12.1 Exclusion of full-flag punctures

Let $P \subset E$ be the inverse image of W. Then $\mathrm{rk}\, P = 6$ and $\deg P = -1$. At a full-flag puncture, write the cyclic gaps as $x, y, z > 0$ with $x + y + z = 1$. The adjoint weights are two zeros and

$$x, y, z, 1 - x, 1 - y, 1 - z.$$

Any rank-six induced fibre has weight at least the sum of the four smallest root weights. The six root weights total three, while each of the two largest is strictly less than one, so the four-smallest sum is > 1. Therefore $\mathrm{pardeg}(P) > 0$, contradicting parabolic stability. Every non-scalar puncture in this branch is of type $2 + 1$.

12.2 Universal-extension obstruction

Let $P \subset E$ be the inverse image of $W = \mathcal{O}_C^5$. Parabolic stability gives $H^0(P) = 0$, and the branch data give $h^0(Q \otimes L) \leq 1$. Proposition 6.4, with $g = 5$ and $h = 1$, yields

$$\mathcal{O}_C^4 \subset \ker \beta.$$

Parabolic stability also gives $H^0(C, E) = 0$: a nonzero holomorphic section would generate a saturated parabolic line of nonnegative parabolic degree inside the stable parabolic degree-zero bundle E. Thus the full boundary map $H^0(C, G) \to H^1(C, M)$ is injective, and Proposition 6.3 contradicts the displayed rank-four trivial subbundle. Hence the final high-HN branch is impossible.

13 Independent canonical-Deligne determinant reconstruction

The preceding boundary-cocycle obstruction makes the determinant character unnecessary for the main proof. We retain its integral construction and its second elimination of the former minimal-nilpotent endpoint as an independent reconstruction.

At a $2+1$ puncture, a scalar twist lets us choose canonical residue

$$\mathrm{diag}(0, 0, \delta), \qquad 0 < \delta < 1.$$

The adjoint residues on E_{13}, E_{23} are $-\delta$, those on E_{31}, E_{32} are δ, and the Levi part has residue zero. Moving the negative residues into $[0, 1)$ multiplies precisely E_{13}, E_{23} by z. Thus, with $R = \mathbf{C}[[z]]$ and $K = \mathbf{C}((z))$, the canonical adjoint lattice is

$$\mathfrak{p}_R = \left\{ \begin{pmatrix} A & zb \\ c & d \end{pmatrix} : \mathrm{tr}\, A + d = 0 \right\} \subset \mathfrak{sl}_3(K).$$

Equivalently, it is the trace-zero endomorphism lattice of the periodic chain

$$z\Lambda_0 \subset \Lambda_1 \subset \Lambda_0, \qquad \Lambda_0 = R^3, \quad \Lambda_1 = zRe_1 \oplus zRe_2 \oplus Re_3.$$

At a scalar puncture the chain is hyperspecial. The following theorem is stated for an arbitrary periodic chain because no contact-order or orbit reduction is needed.

Theorem 13.1 (Integral minimal-nilpotent determinant morphism)**.** *Let $\Lambda_\bullet$ be a periodic lattice chain in an r-dimensional K-vector space and put*

$$\mathfrak{p}_R = \mathrm{End}_{\mathrm{ch}}(\Lambda_\bullet) \cap \mathfrak{sl}_r(K).$$

Let $\mathcal{M} \subset \mathfrak{p}_R$ be a saturated rank-one module whose generic fibre is a minimal-nilpotent line. If

$$\mathcal{K} = (N_{\mathfrak{sl}_r(K)}(K\mathcal{M}) \cap \mathfrak{p}_R)/\mathcal{M},$$

then there is a holomorphic, generically nonzero morphism

$$\det \mathcal{K} \longrightarrow \mathcal{M}^{\otimes(r-2)}.$$

It may have zeros at the boundary but cannot have a pole.

Proof. Let

$$I_K = \operatorname{im} m, \qquad H_K = \ker m, \qquad J_K = H_K/I_K, \qquad Q_K = V_K/H_K.$$

The induced intersections and quotients of the periodic chain give free chains $I_\bullet, J_\bullet, Q_\bullet$. A field map in the generic minimal-nilpotent line preserves the original chain exactly when the induced map $Q_K \to I_K$ preserves the quotient chains. Saturation therefore gives the equality of integral lattices

$$\mathcal{M} = \operatorname{Hom}_{\mathrm{ch}}(Q_\bullet, I_\bullet). \tag{6}$$

Let $\mathcal{N} = N_{\mathfrak{sl}_r(K)}(K\mathcal{M}) \cap \mathfrak{p}_R$. The field normalizer preserves the flag $I_K \subset H_K \subset V_K$, hence $\mathcal{N}$ acts on the three induced chains. Let $\mathcal{D}$ be its image on the graded chains and $\mathcal{U}$ its kernel. There is an exact sequence

$$0 \longrightarrow \mathcal{U}/\mathcal{M} \longrightarrow \mathcal{K} \longrightarrow \mathcal{D} \longrightarrow 0.$$

The two nontrivial first off-diagonal blocks give an integral injection, generically an isomorphism,

$$\mathcal{U}/\mathcal{M} \longrightarrow A \oplus B, \qquad A = \operatorname{Hom}_{\mathrm{ch}}(J_\bullet, I_\bullet), \quad B = \operatorname{Hom}_{\mathrm{ch}}(Q_\bullet, J_\bullet).$$

If both blocks vanish, the endomorphism factors from $Q_\bullet$ to $I_\bullet$ and belongs to $\mathcal{M}$ by (6). Thus

$$\det(\mathcal{U}/\mathcal{M}) \longrightarrow \det A \otimes \det B$$

is integral and generically nonzero.

It remains to treat the diagonal factor without choosing a generic volume. Put

$$\mathcal{B} = \operatorname{End}_{\mathrm{ch}}(I_\bullet) \oplus \operatorname{End}_{\mathrm{ch}}(J_\bullet) \oplus \operatorname{End}_{\mathrm{ch}}(Q_\bullet).$$

The trace map

$$\tau : \mathcal{B} \longrightarrow R, \qquad (a, A, c) \longmapsto a + \operatorname{tr}(A) + c,$$

is split surjective; its kernel is the integral trace-zero graded module $\mathcal{D}_0$. Hence

$$\det \mathcal{D}_0 \simeq \det \mathcal{B} \simeq \det \operatorname{End}_{\mathrm{ch}}(J_\bullet).$$

Choose one lattice J_0 in the chain. Inclusion into $\operatorname{End}_R(J_0)$ and the canonical identity $\det \operatorname{End}_R(J_0) \simeq R$ give an integral map

$$\det \mathcal{D}_0 \longrightarrow R.$$

Since $\mathcal{D}$ is a full-rank sublattice of $\mathcal{D}_0$, we obtain $\det \mathcal{D} \to R$, again with zeros but no poles.

Finally, composition of chain maps is integral:

$$A \otimes B \longrightarrow \mathcal{M}.$$

Since $\operatorname{rk} J_K = r - 2$, the determinant of this perfect generic pairing gives

$$\det A \otimes \det B \longrightarrow \mathcal{M}^{\otimes(r-2)}.$$

Combining the three integral determinant maps proves the theorem. $\square$

Corollary 13.2 (Global no-pole determinant character)**.** *Let E be a canonical adjoint Deligne bundle on a smooth complete curve, with its periodic-chain lattices at the punctures. Let $M \subset E$ be a saturated line with generically minimal-nilpotent fibre, and let K be its global quotient normalizer. Then the intrinsic generic determinant character extends to a generically nonzero morphism*

$$\det K \longrightarrow M^{\otimes(r-2)}$$

on the whole curve.

Proof. Over the function field, the flag $I = \operatorname{im} M \subset H = \ker M \subset V$ and the composition pairing

$$\operatorname{Hom}(H/I, I) \otimes \operatorname{Hom}(V/H, H/I) \longrightarrow \operatorname{Hom}(V/H, I) = M$$

define one intrinsic determinant character. The determinant trivialization of the trace-zero diagonal factor is likewise functorial; equivalently, it is the canonical identity $\det \operatorname{End}(H/I) \simeq \mathcal{O}$. Thus every completed-DVR construction in theorem 13.1 restricts to this same generic morphism, rather than to unrelated local choices. The theorem says that this rational morphism has no pole at any closed point. A rational section of a line bundle on a smooth curve that is regular at every completed DVR is global, proving the claim. $\square$

Proposition 13.3 (Local–global kernel identity)**.** *Let E be the canonical adjoint Deligne bundle on a smooth curve, let $M \subset E$ be saturated, and put $G = E/M$. If*

$$\beta : G \longrightarrow G \otimes M^{-1}$$

is the quotient bracket and $\mathcal{K}_{\mathrm{glob}} = \ker \beta$, fix a puncture with completed local ring R, fraction field K, canonical lattice $\mathfrak{p}_R$, and completed line $\widehat{M}$. Then the completed stalk is exactly

$$\left(N_{\mathfrak{sl}_r(K)}(K\widehat{M}) \cap \mathfrak{p}_R\right) / \widehat{M}.$$

Proof. The image of β is torsion-free, so the kernel is saturated and locally free. Completion is flat and commutes with the kernel. Choose a generator m of the free rank-one R-module $\widehat{M}$. For $x \in \mathfrak{p}_R$,

$$\bar{x} \in \ker \beta_R \iff [m, x] \in Rm.$$

The canonical chain-endomorphism lattice is closed under bracket. Since the global line is saturated, completion remains saturated and

$$(K\widehat{M}) \cap \mathfrak{p}_R = \widehat{M},$$

so the preceding condition is equivalent to $[m, x] \in Km$, i.e. to $x \in N_{\mathfrak{sl}_r(K)}(Km) \cap \mathfrak{p}_R$. $\square$

In the final high-HN branch, $r = 3$ and the universal-extension argument gives $\mathcal{O}_C^4 \subset \mathcal{K}_{\mathrm{glob}}$, generically an equality. Taking determinants and applying theorems 13.2 and 13.3 gives

$$\mathcal{O}_C \longrightarrow \det \mathcal{K}_{\mathrm{glob}} \longrightarrow M.$$

This is impossible because $\deg M = -1$. Thus the determinant argument independently recovers the elimination of the final high-HN branch for arbitrary contact order.

14 Propagation to all rank-three representations

For an irreducible unitary rank-three system, the preceding arguments supply

$$H^0(B, R^1\pi_* \operatorname{ad} \mathcal{V}) = 0$$

on every finite etale moduli cover B in every genus $g \geq 5$. The smaller actual adjoint constituents lie in the published strict range or in the endpoint lemma. The Leray argument of [4, Proposition 8.2.1] gives strong cohomological rigidity.

The formal constancy step needs vanishing on the dominant etale versal base used for a linear lift. The following proposition separates that issue from the finite-cover geometry.

Proposition 14.1 (Projective finite-index descent)**.** *Let $B \to \mathcal{M}_{g,n}$ be a connected finite etale scheme cover carrying the canonical projective globalization of an irreducible MCG-finite rank-three representation, and let*

$$\mathcal{U}_B = \mathrm{ad}(\mathbb{P}\rho_B)$$

be its honest adjoint local system. Suppose that for every connected finite etale $B' \to B$,

$$H^0(B', R^1\pi_*\mathcal{U}_{B'}) = 0. \tag{14.1.1}$$

Let $M \to \mathcal{M}_{g,n}$ be connected, dominant and etale, and let γ_0 be a complex local system on the punctured total curve whose fibral restriction is the same irreducible representation. Then

$$H^0(M, R^1\pi_* \,\mathrm{ad}\,\gamma_0) = 0.$$

Proof. Let X_0 be the connected component of $M \times_{\mathcal{M}_{g,n}} B$ containing the chosen matching lift. It is surjective finite etale over M and dominant etale over B; a nonzero flat section remains nonzero after pullback. After one fibral conjugation, the linear local system $\gamma_0|_{X_0}$ restricts exactly to the chosen irreducible fibre representation ρ. For an element x of the total fundamental group, $\gamma_0(x)$ is therefore an exact linear intertwiner from ρ to ρ^x. By the construction in [4, Lemma 2.2.2], the canonical projective globalization assigns to x the projective class of another exact linear intertwiner M_x for the same pair. Thus $M_x^{-1}\gamma_0(x)$ commutes with ρ and is scalar by Schur's lemma. Consequently the projective classes agree and

$$\mathrm{ad}\,\gamma_0 \simeq f^*\mathcal{U}_B.$$

The use of exact linear intertwiners is essential: arbitrary projective extensions may differ by a finite self-twist, which can alternatively be killed after a further finite etale cover. The image

$$\Gamma = \mathrm{im}(\pi_1(X_0) \to \pi_1(B))$$

has finite index by [4, Lemma 2.1.4]. A global section is therefore a vector fixed by Γ. It becomes a nonzero global section on the connected finite topological cover corresponding to Γ, which is a finite etale algebraic cover $B_\Gamma \to B$ by Riemann existence, contradicting (14.1.1). ☐

Lemma 14.2 (Direct Artin devissage)**.** *Let A be a local Artin $\mathbf{C}$-algebra and let $\mathcal{W}_A$ be a local system of free A-modules on a punctured family. Put $\mathcal{W}_0 = \mathcal{W}_A \otimes_A \mathbf{C}$. If*

$$R^0\pi_*\mathcal{W}_0 = 0, \qquad H^0(M, R^1\pi_*\mathcal{W}_0) = 0,$$

then

$$R^0\pi_*\mathcal{W}_A = 0, \qquad H^0(M, R^1\pi_*\mathcal{W}_A) = 0.$$

Proof. Induct on the length of A. For a small quotient

$$0 \to I \to A \to A' \to 0, \qquad \mathfrak{m}_A I = 0,$$

freeness gives

$$0 \to I \otimes_{\mathbf{C}} \mathcal{W}_0 \to \mathcal{W}_A \to \mathcal{W}_{A'} \to 0.$$

The action on the kernel is residual. The long exact sequence of higher direct images and induction first give $R^0\pi_*\mathcal{W}_A = 0$, and then an exact segment

$$0 \to I \otimes R^1\pi_*\mathcal{W}_0 \to R^1\pi_*\mathcal{W}_A \to R^1\pi_*\mathcal{W}_{A'}.$$

Taking global sections proves the claim. ☐

The preceding argument applies only when the residual unitary fibre is irreducible. This qualification is essential: [4, Lemma 8.5.1] assumes only unitarity, and the deformation to a PVHS need not preserve irreducibility. We now treat every residual rank-three type.

Lemma 14.3 (Exact trivial-coefficient gap)**.** *Let $g \geq 2$. Every nonzero sub-local system of $R^1\pi_*^\circ \mathbf{C}$ has rank at least $2g$.*

Proof. Apply [4, Lemma 6.1.1] to an irreducible sub-local system L. For the trivial coefficient, $E = \mathcal{O}_C$ and every nonzero $v \in H^0(\omega_C(D))$ acts injectively on $H^0(\omega_C)$, so the multiplication rank is g. Lemma 6.1.1 bounds it above by $\operatorname{rk} L/2$. Therefore $\operatorname{rk} L \geq 2g$. ☐

Lemma 14.4 (Reducible rank-three adjoint multiplicities)**.** *Let R be semisimple, reducible, non-scalar, and of rank three. In an isotypic decomposition*

$$\operatorname{ad} R \simeq \bigoplus_i U_i^{\oplus w_i}, \qquad u_i = \operatorname{rk} U_i,$$

the possible multiplicities satisfy

type of R	*possible (u_i, w_i)*
$1+1+1$, *not scalar*	$(1, w \leq 4)$,
$2+1$	$(1, w \leq 3), (2, w \leq 3), (3, 1)$.

In particular, for $g \geq 5$ one has $w_i < 2g - 2u_i$.

Proof. For three characters, $\operatorname{ad} R$ is the two-dimensional diagonal part together with the six ratios $\chi_i\chi_j^{-1}$. If two characters agree, the trivial coefficient has multiplicity four and each cross character has multiplicity two; if all three are distinct, every multiplicity is at most three.

For $R = T \oplus \chi$, with T irreducible of rank two,

$$\operatorname{ad} R \simeq \mathbf{1} \oplus \operatorname{ad} T \oplus (T \otimes \chi^{-1}) \oplus (T^\vee \otimes \chi).$$

A rank-one constituent can occur at most three times inside $\operatorname{ad} T$; a rank-two constituent can occur in the two cross terms and at most once in $\operatorname{ad} T$; and a rank-three constituent occurs at most once. At genus five the corresponding fixed-part lower bounds are $8, 6, 4$, respectively, and the inequalities improve with genus. ☐

Proposition 14.5 (Rank-three formal propagation)**.** *Assume $g \geq 5$, and assume the finite-cover adjoint fixed-part vanishing proved above for every irreducible unitary MCG-finite rank-three fibre. Let γ be any free Artin deformation on a dominant etale versal base whose fibral restriction is the constant deformation of a unitary rank-three system R. Then*

$$H^0(M, R^1\pi_*^\circ \operatorname{ad} \gamma) = 0.$$

Proof. If R is irreducible, Schur gives $R^0\pi_*^\circ \operatorname{ad} R = 0$, Proposition 14.1 gives residual fixed-part vanishing, and Lemma 14.2 propagates it through every small extension.

Suppose R is reducible and non-scalar. After the dominant etale base change of [4, Lemma 2.4.2], write

$$\operatorname{ad} \gamma \simeq \bigoplus_i U_i \otimes \pi^* W_i,$$

where W_i is free of rank w_i. The maximal-ideal filtration on W_i has graded pieces that are direct sums of copies of $W_i^0 = W_i \otimes_A \mathbf{C}$. A first nonzero invariant therefore produces a nonzero complex map

$$(W_i^0)^\vee \longrightarrow R^1\pi_*^\circ U_i$$

whose image has rank at most w_i. Lemma 14.4 gives $w_i < 2g - 2u_i$, contradicting [4, Theorem 1.7.1].

Finally suppose $R = \chi^{\oplus 3}$. Then $\operatorname{ad} R$ is eight copies of the trivial fibral coefficient. In the decomposition of [4, Lemma 2.4.2], any rank-one unitary factor that is trivial on the fibres is pulled back from the base and is absorbed into the coefficient module. Write the result as $\operatorname{ad}\gamma \simeq \pi^* W$, where W is free of rank eight, and put $W^0 = W \otimes_A \mathbf{C}$. A first nonzero Artin graded piece then gives a nonzero map

$$(W^0)^\vee \longrightarrow R^1\pi^\circ_* \mathbf{C}$$

whose image is a sub-local system of rank at most eight. This contradicts Lemma 14.3, since $8 < 2g$ for $g \geq 5$. $\square$

Proposition 14.5 supplies exactly the hypothesis of [4, Proposition 3.2.1] for every Artin truncation. It does not assume that a finite-cover geometric contradiction persists on an arbitrary dominant etale base.

Published propagation map. For clarity, the remaining uses of the strict Landesman–Litt rank bound are replaced as follows. In the proof of [4, Proposition 8.2.1], the hypothesis $r < \sqrt{g+1}$ is used to get $\operatorname{rk}\operatorname{ad}(V) = r^2 - 1 < g$ and invoke [4, Theorem 6.2.1]; the fixed-part vanishing and finite-index descent proved above supply that input here. The proofs of [4, Lemmas 8.3.3 and 8.3.4] then use strong rigidity together with irreducibility, $g \geq 3$, quasi-unipotent boundary monodromy, and finite determinant, but no further rank estimate. The isomonodromy input used in [4, Proposition 8.4.1] has the larger range $r < 2\sqrt{g+1}$, which holds for $r = 3$ and $g \geq 5$.

In the proof of [4, Lemma 8.5.1], the strict bound is again used only to obtain $\operatorname{rk}\operatorname{ad}(\gamma) = r^2 - 1 < g$ and apply [4, Theorem 6.2.1] at every Artin truncation; Proposition 14.5 replaces that application. The proof of [4, Lemma 8.5.2] uses the rank bound only through Lemma 8.5.1. In the proof of [4, Lemma 8.6.1], the original bound enters in the estimate

$$n_i \dim \sigma_i \leq \dim \rho_1 \dim \rho_2 < \frac{g+1}{4}$$

before [4, Theorem 7.2.1] is applied. Here total rank at most three instead gives $n_i \dim \sigma_i \leq 2$, while the same published lower bound is at least six. Proposition 14.8 records that replacement, and no further numerical bound enters the socle induction of [4, Section 8.7].

Proposition 14.6 (Unitary rank-three finite image)**.** *Let $g \geq 5$. Every unitary MCG-finite rank-three representation has finite image.*

Proof. If the representation is reducible, its irreducible summands have rank at most two and have finite image by the published strict theorem. Assume it is irreducible. If its projective image is finite, then its image is finite: the scalar kernel has finite image because the determinant has finite image, and the kernel of $z \mapsto z^3$ is finite. We may therefore assume the projective image is infinite. The specialized adjoint fixed-part vanishing and the Leray argument of [4, Proposition 8.2.1] give strong cohomological rigidity. The proofs of [4, Lemmas 8.3.3 and 8.3.4] use the numerical rank hypothesis only through that rigidity; their remaining inputs are $g \geq 3$, finite determinant, and quasi-unipotent boundary monodromy. Thus the representation is integral.

Strong rigidity is preserved under every complex embedding of its number field of definition. Lemma [4, Lemma 4.3.2] makes every conjugate total-space representation a complex PVHS. The isomonodromy input used in [4, Proposition 8.4.1] applies in the larger range $3 < 2\sqrt{g+1}$, automatic here, so every conjugate fibre representation is unitary. In the current arXiv version of [5] this input is Theorem 1.2.13; [4] cites an earlier numbering, Theorem 1.2.12. The arithmetic unitary criterion at the end of that proposition then gives finite image. □

Proposition 14.7 (Semisimple rank-three finite image)**.** *Let $g \geq 5$. Every semisimple MCG-finite rank-three representation has finite image.*

Proof. The reducible case has rank-one and rank-two summands and is published. For an irreducible system, use the finite-determinant linear globalization and the constant-determinant nonabelian-Hodge deformation of [4, Theorem 4.3.1]. Its PVHS endpoint is unitary because $3 < 2\sqrt{g+1}$ and is MCG-finite by [4, Proposition 2.1.3]. Proposition 14.6 makes the endpoint finite. Proposition 14.5 is precisely the replacement for [4, Theorem 6.2.1] in [4, Lemma 8.5.1]; hence the formal restriction is constant. In the proof of [4, Lemma 8.5.2], the numerical rank hypothesis is used only through Lemma 8.5.1. The remaining argument spreads formal constancy to a dense open and then uses closedness of the semisimple conjugacy orbit. It therefore shows that the original fibre is conjugate to the unitary endpoint. □

Proposition 14.8 (Rank-three characteristic extensions)**.** *Let $g \geq 5$. An MCG-finite extension of two semisimple finite-image representations of total rank at most three has finite image, and hence splits, provided the subobject is characteristic.*

Proof. Let the factor ranks be a, b. Then $a + b \leq 3$, so $ab \leq 2$. Decompose

$$\rho_2^\vee \otimes \rho_1 = \bigoplus_i \sigma_i^{\oplus n_i}, \qquad d_i = \dim \sigma_i.$$

Thus $n_i d_i \leq 2$ and $d_i \leq 2$. Pass to a finite Galois cover $\Sigma'_{g',n'} \to \Sigma_{g,n}$ on which both factors are trivial. The one-step unipotent restriction factors through an equivariant homology map

$$H_1(\Sigma'_{g',n'}; \mathbf{C}) \longrightarrow \rho_2^\vee \otimes \rho_1.$$

Because the subrepresentation is characteristic, the argument of [4, Lemma 8.6.1] makes each projected isotypic quotient stable under a finite-index mapping-class subgroup. Its dimension is at most $n_i d_i \leq 2$. Equivariant duality gives a stable subrepresentation of the same dimension. By [4, Theorem 7.2.1], every nonzero such subrepresentation has dimension at least

$$2g - 2d_i \geq 6.$$

Hence every projected map vanishes and the restriction to the finite cover is trivial. Thus the original image is finite; over $\mathbf{C}$ the extension then splits. □

The semisimplification of an arbitrary MCG-finite rank-three representation is MCG-finite and finite by Proposition 14.7. Applying Proposition 14.8 inductively to the characteristic socle filtration, exactly as in [4, Section 8.7], proves theorem 1.1.

Status of the argument

The theorem is new and unrefereed. The endpoint and coparabolic convention, the genus-five branch reconstruction, the normalizer boundary-cocycle obstruction, and the formal-propagation interfaces have each been re-derived within the project. The spectral-projector closure and periodic-chain determinant theorem are retained as separate reconstructions, not as load-bearing inputs. The remaining external task is specialist verification of the assembled use of the Landesman–Litt Hodge, parabolic, and deformation machinery. Exact scripts used during development verify only finite arithmetic and fibrewise linear algebra; they are not substitutes for those geometric inputs.

AI-assistance disclosure

OpenAI GPT-5.6 Pro and GPT-5.6 Sol were used for drafting, symbolic exploration, generation of finite verification scripts, and checks of successive versions for omitted cases and mismatched hypotheses. The author directed their use and is responsible for the manuscript and any errors. These tools were not treated as mathematical authorities, and the finite scripts do not verify the imported geometric inputs.

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